

Tutorial 6_summary

Tutorial 6 — DC Bridge Analysis, Linearization & Signal Amplification

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1. Basic DC Bridge Analysis

1.1 Why a Bridge?

Sensors (e.g., RTDs, strain gauges) typically output a **minute change in resistance** — instruments cannot read this directly. The **Deflection Bridge** is a conversion circuit that transforms ΔR into a **measurable voltage difference**.

1.2 Key Parameters

Symbol	Name	Role
V_S	Supply voltage	Drives the system; determines upper output limit and power consumption
E_{Th}	Output voltage	Thévenin equivalent potential at measurement terminals
R_1 (or R_T)	Active bridge arm	The sensor itself — resistance changes with measurand
R_2, R_3, R_4	Fixed bridge arms	Precision reference resistors selected by the engineer

1.3 Three Core Formulas

1) Output Voltage Formula:

$$E_{Th} = V_S \left(\frac{R_1}{R_1 + R_4} - \frac{R_2}{R_2 + R_3} \right)$$

2) Bridge Balance Condition ($E_{Th} = 0$):

$$\frac{R_4}{R_1} = \frac{R_3}{R_2}$$

3) RTD Temperature Characteristic (Metal Resistance):

$$R_T = R_0(1 + \alpha T)$$

where R_0 = resistance at 0 °C, α = temperature coefficient.

2. Bridge Design — Worked Example

Problem Setup

- RTD sensor: $\alpha = 0.00385/^\circ\text{C}$, $R_0 = 100\ \Omega$
- Bridge arms: $R_2 = R_3 = R_4 = 100\ \Omega$
- Supply voltage $V_S = 10\ \text{V}$

(a) Output at 0 °C and 100 °C

At 0 °C: $R_1 = R_0 = 100\ \Omega$, bridge balanced $\rightarrow E_{Th} = 0\ \text{V}$

At 100 °C:

$$R_1 = 100(1 + 0.00385 \times 100) = 138.5\ \Omega$$

$$E_{Th} = 10 \left(\frac{138.5}{138.5 + 100} - \frac{100}{100 + 100} \right) = 10(0.5808 - 0.5) = 0.808\ \text{V}$$

(b) Design V_S to output 100 mV at 100 °C

Scaling proportionally: $V_S = 10 \times \frac{0.1}{0.808} = 1.24\ \text{V}$

3. Engineering Design and Linearization of Sensor Bridges

3.1 The Core Pain Point

State	Description
Ideal	Perfectly proportional, linear relationship between measurand and V_{out}
Reality	Bridge voltage division has inherent nonlinearity ; some sensors (thermistors) exhibit severe exponential nonlinearity

Engineering Goal: Without software, use clever hardware circuit parameter choices to "force" the output curve into a straight line.

3.2 Strategy One: Large Bridge Arm Ratio Method

For: Small ΔR sensors like RTD platinum resistors.

Design Concept: Make fixed bridge arm resistance much larger than sensor resistance. Bridge ratio $r = R_3/R_2 \gg 1$. Use a massive fixed base value to "dilute/mask" fluctuations in the denominator.

$$V_{OUT} \approx V_S \cdot \alpha T \cdot \frac{1}{r}$$

Trade-off	Effect
Linearity	✅ Excellent — denominator fluctuations are negligible
Sensitivity	❌ Reduced — output voltage is smaller

3.3 Strategy Two: Three-Point Design Method

For: Severe nonlinearity sensors like **thermistors** ($R_\theta \propto e^{1/T}$).

Design Concept: Use the "upward curvature" of the bridge output curve to **precisely cancel** the "downward curvature" from the exponential drop in thermistor resistance.

Three-Step Guide:

1. **Lock in** Start Point, Midpoint, and End Point of the measurement temperature range
2. **Mandate** that these three points output perfectly equidistant voltages (e.g., 0 V, 0.5 V, 1.0 V)
3. **Set up** a system of 3 equations → solve for the unique Bridge Ratio r and Supply Voltage V_S

✅ For thermistors, the optimal r is typically between **0.25 ~ 0.30**.

4. System-Level Integration — Bridge + Amplifier

4.1 Why an Amplifier?

Reality	Requirement	Gap
Bridge output E_{Th} : few mV → μV	ADC/microcontroller needs: 1.0 V ~ 5.0 V	×100 to ×1000

4.2 The Ideal Op-Amp — Two "Magic Tricks"

Assumption	Meaning	Consequence
Virtual Open $i_+ = i_- = 0$	Infinite input impedance	Never "steals" current from the bridge
Virtual Short $V_+ = V_-$	Under negative feedback, both inputs forced to same potential	Simplifies analysis

✅ Virtual Open guarantees that the op-amp **doesn't load** the bridge — all bridge formulas remain valid.

4.3 Ultimate Formula — Full Bridge + Differential Amplifier

For a **four-arm active bridge** (all 4 strain gauges) connected to a differential amplifier:

$$V_{OUT} = 2 \cdot V_S \cdot G \cdot e \cdot \frac{R_F}{R_0}$$

Symbol	Meaning	Example
V_{OUT}	Final standard output voltage	1.0 V
R_F	Feedback resistor (only component to calculate!)	To be designed
R_0	Initial sensor resistance	120 Ω
V_S	Bridge supply voltage	15 V
G	Gauge factor of strain gauge	~ 2
e	Physical strain	—

✔ **Only R_F needs to be calculated** — the rest are sensor specs or system requirements!

5. Summary Table

Topic	Key Formula	Notes
Bridge output	$E_{Th} = V_S \left(\frac{R_1}{R_1 + R_4} - \frac{R_2}{R_2 + R_3} \right)$	General formula
Balance condition	$R_4 / R_1 = R_3 / R_2$	Required for $E_{Th} = 0$
RTD $R(T)$	$R_T = R_0(1 + \alpha T)$	Linear approximation (metal)
Large ratio approx	$V_{OUT} \approx V_S \alpha T / r$	Sacrifice sensitivity for linearity
Three-point method	Solve 3 equations for (r, V_S)	For thermistor linearization
Full bridge + diff amp	$V_{OUT} = 2V_S G e (R_F / R_0)$	Complete system design

✔ **Key Takeaway:** The Wheatstone bridge is the fundamental interface between resistive sensors and the electrical world. Bridge **balance** ($R_4 / R_1 = R_3 / R_2$) zeros the output at reference. The **Large Ratio Method** ($r \gg 1$) and **Three-Point Design Method** are the two engineering strategies for linearization. Cascading a **differential amplifier** amplifies the μV – mV bridge output to standard 1–5 V levels — with the **feedback resistor** R_F being the sole design parameter.